

NEXA ENTERPRISE DATA PLATFORM | TECHNICAL ARCHITECTURE SPECIFICATION

Synapse Real-Time Order & Telemetry ETL Pipeline

Target Warehouse: Meridian Snowflake Data Lakehouse | Engine: Apache Spark + dbt Core

Status: PRODUCTION APPROVED

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SLA Target: P99 < 180s

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Source Systems:	Kafka CDC, Postgres, S3	Orchestrator:	Prefect Enterprise	Security Level:	Tier-1 Confidential / PII

1. Executive Overview & System Objectives

The **Synapse ETL Pipeline** ingests high-frequency transactional order streams and device telemetry logs across global regional hubs into the **Meridian Data Warehouse**. It powers operational dashboards, fraud detection models, and financial compliance reporting for Nexa Commerce and Vertex Cloud Services.

DAILY EVENT VOLUME

450 Million

~ 280 GB Compressed Delta / Day

END-TO-END LATENCY

P99 < 180 Sec

CDC Stream to Gold Warehouse

DATA AVAILABILITY

99.99% SLA

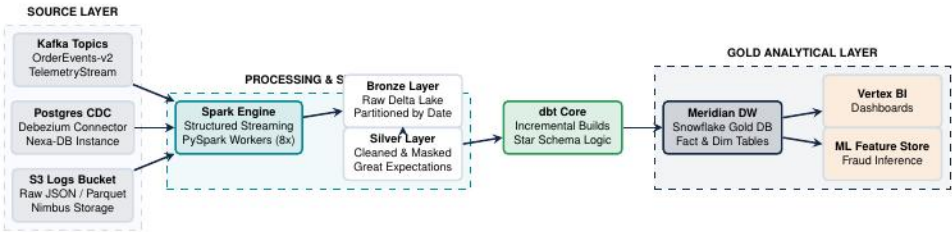
Zero Data Loss / DLO Enabled

1.1 Key Design Principles

- Medallion Architecture Alignment:** Strict separation between Raw (Bronze), Cleansed/Enriched (Silver), and Business Dimensional (Gold) layers.
- Idempotent Stream Processing:** All ingestion handlers enforce exact-once delivery semantics via deterministic message key hashing and Snowflake MERGE operations.
- Automated Schema Evolution:** Schema changes in source Kafka topics are parsed through the Vertex Schema Registry, enforcing backward-compatible migrations.

2. End-to-End Pipeline Architecture

The pipeline topology transitions real-time events from edge data sources through structured streaming layers into analytical gold tables.



3. Data Warehouse Schema & Dimensional Specification

The target analytical layer in **Meridian Snowflake Warehouse** adopts a Kimball Star Schema design optimized for high-throughput aggregation and fast analytical query processing.

3.1 Gold Fact Table: gold_fact_orders

This table stores transactional order activity enriched with currency conversion and dimension keys.